

There are some oscilloscopes having rich features but are available for just over US\$ 300. For example, a budget-friendly analogue oscilloscope such as the one shown in Fig. 4 is a modern 30MHz digital readout oscilloscope with built-in component tester, computer interface (option), colour LCD readout and other built-in options (at extra cost) like function generator, curve tracer, logic scope, power supply, frequency counter or digital voltmeter.

Amidst a plethora of brands and specifications available in the market, and with a limited budget, getting the right oscilloscope is a daunting task. Let us explore some important aspects in low-cost, budget-friendly oscilloscopes.

Features to look for

"Budget-friendly oscilloscopes are basic scopes with limited functionalities and performances for very cost-conscious customers. These scopes have no rich features but can be used for simple testing and measurement," says April Fang, sales manager, Siglent Technologies Co. Ltd, Shenzhen. While this is true, it is always good to look out for better options that meet your exact requirement at minimal cost.

There are many vendors supplying low-cost oscilloscopes that have more functions and upgradable features. "In budget-friendly scopes one has to look for good performance combined with functionality, form factor and battery operation. Performance includes higher sampling rate with larger memory to provide better analysis of the signals. A higher waveform-acquisition rate enables faster capture of faults to reduce debug time. The product has to offer upgradable bandwidth and serial protocol decoding features to cater to future requirements of the customer to provide investment protection," says Srinivasa Appalla, manager - product support and application, Rohde & Schwarz India Pvt Ltd.



Fig. 3: Typical PC based oscilloscope



Fig. 4: Typical analogue oscilloscope (benchtop)

Types of oscilloscope

- Analogue oscilloscopes
- Digital storage oscilloscopes
- Sampling oscilloscopes
- Mixed-signal oscilloscopes
- PC based oscilloscopes

Most budget-friendly oscilloscopes are available with sample rate less than or equal to 1GSa/s, bandwidth below 200MHz, 8-bit analogue-to-digital converter, low-waveform capture rate, basic trigger controls, auto measurement and basic mathematical functions for waveform.

With your budget in mind, you should always go for a higher waveform-acquisition rate for faster capture of intermittent or infrequent events like sporadic interference in clock signals. Troubleshooting in the field often requires a variety of functions. The more the functions in an instrument, the less the number of additional instruments required.

"While selecting a budget-friendly oscilloscope one should always keep in mind that the selected oscilloscope should not compromise on signal capture and display. This is important because the primary purpose of an oscilloscope is to capture the signal of interest with the

highest possible fidelity," says Akash Srivastava, application engineer, Keysight Technologies, India.

Some features and specifications you might expect in a scope are given below.

Bandwidth. Bandwidth is one of the most important specifications in an oscilloscope. It specifies the range of frequencies it can reliably measure. Response of the scope will drop off rapidly as the input frequency is raised beyond the specified bandwidth. As a rule of thumb, your scope's bandwidth should be at least five times higher than the fastest digital clock rate in your system under test.

Sampling rate. Sampling defines the number of times a signal is read per second. The faster you sample, the lesser information is lost and the better the scope will represent the waveform or signal on display. It is similar to the frame rate of a movie camera. Sampling rate determines how much waveform detail the scope can capture. But it has some limitations because the faster the sampling rate you have, the faster your memory will be consumed. Therefore a higher sampling rate demands a large memory depth. Scopes with sample rate up to 1GSa/s are easily available in the market.

Memory depth. If you are using the oscilloscope to analyse digital signals, memory depth is extremely important. The larger the memory, the better the scope will capture the signal. However, if the CPU is not capable of handling deep memory records, the process will become slow, lowering the update rate of the scope. Therefore choosing the right memory depth and CPU speed is important. Memory depths of 2Mpts to 14Mpts per channel are available in the market.

Channels. Two or more channel scopes are very common these days. These allow users to read more than one signal at a time, displaying all of these on the screen simultaneously. Each signal read by a scope is fed into a separate channel input.